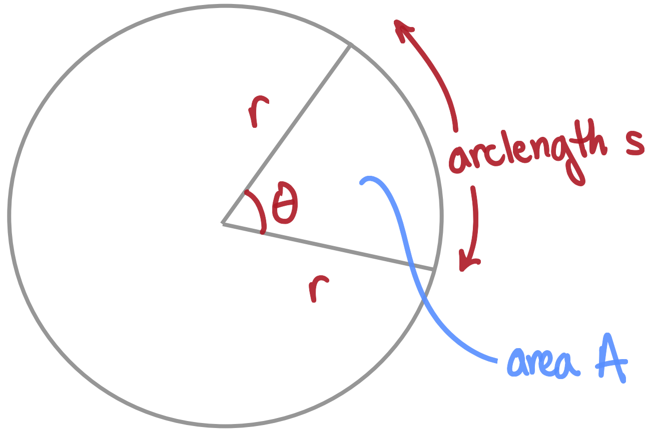
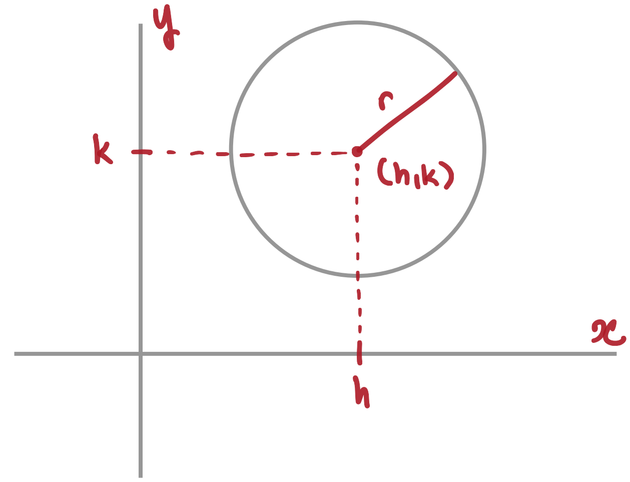




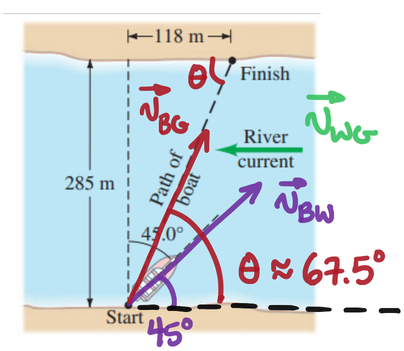
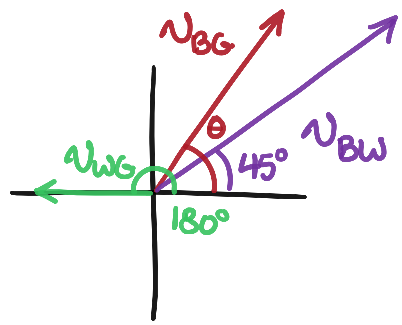
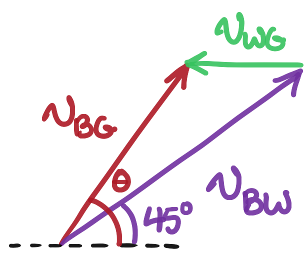
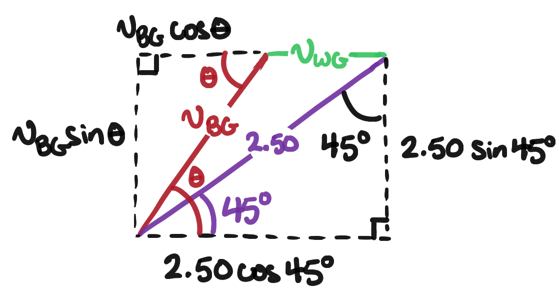
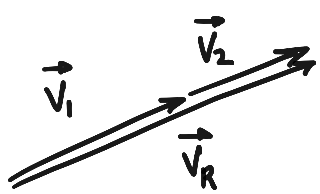


FIGURE 3–45
Problem 48.

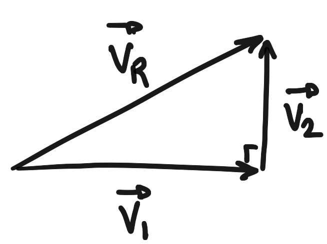




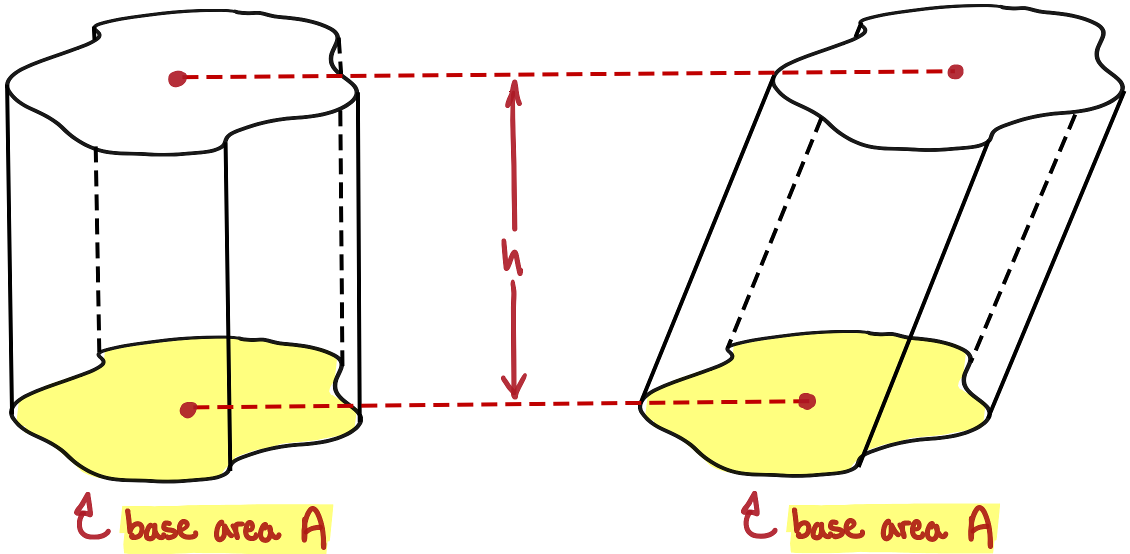


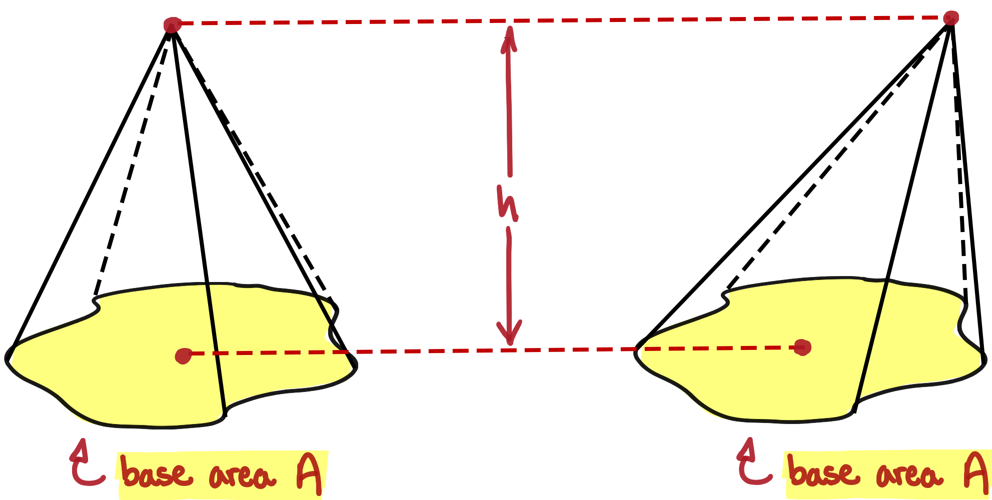


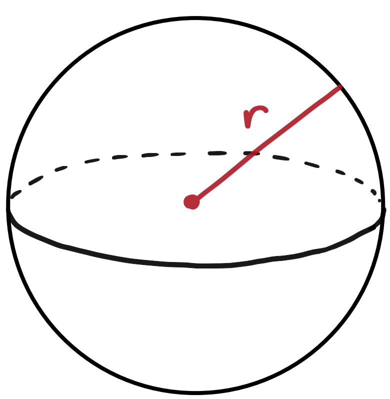
so, $V_1 + V_2 = V_R$

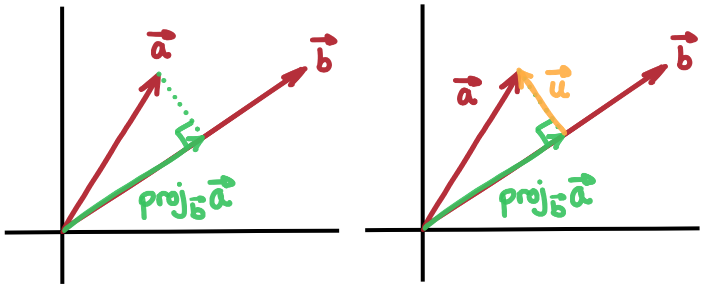


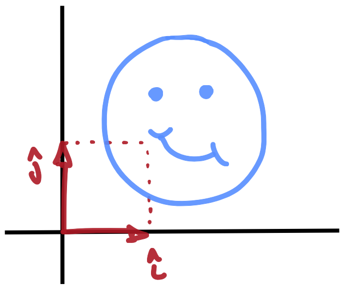
so, $V_1^2 + V_2^2 = V_R^2$


h
base area A
base area A

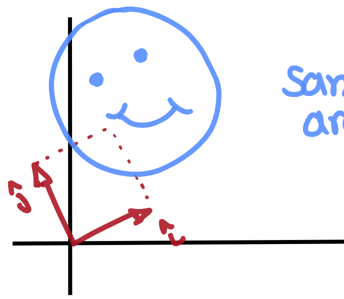

h
base area A
base area A


r





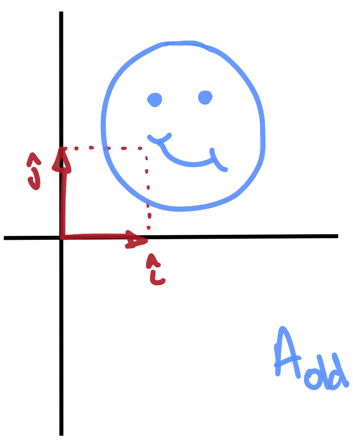
$\xrightarrow[\text{rotation}]{M}$



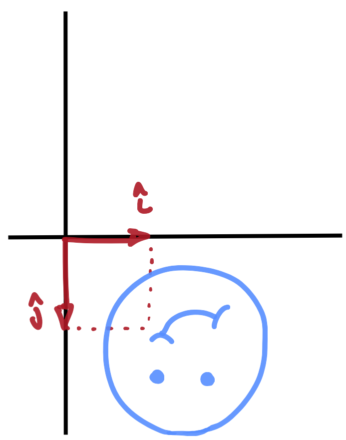
same area

$$A_{old} \cdot \underset{\uparrow}{1} = A_{new}$$

$\det M = 1$



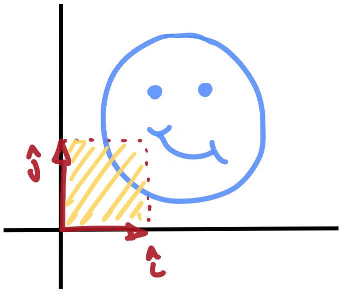
$\xrightarrow[\text{x-axis reflection}]{M}$



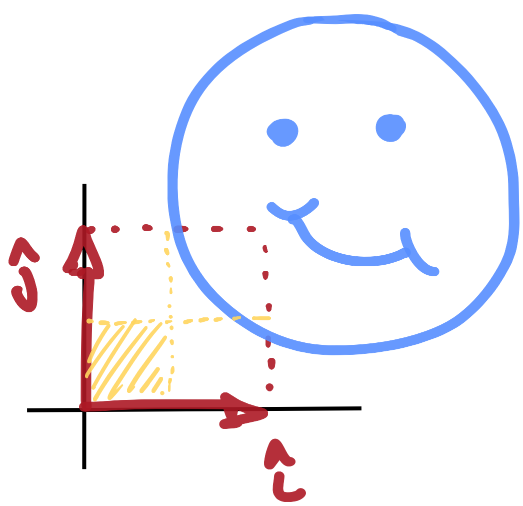
$$A_{old} \cdot \underset{\uparrow}{1} = A_{new}$$

$\det M = -1$

(minus sign indicates flipped image)

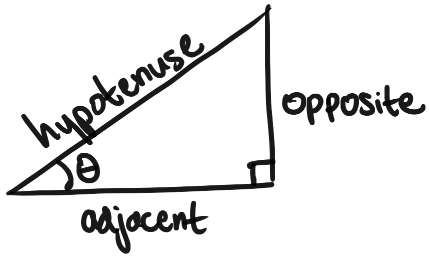


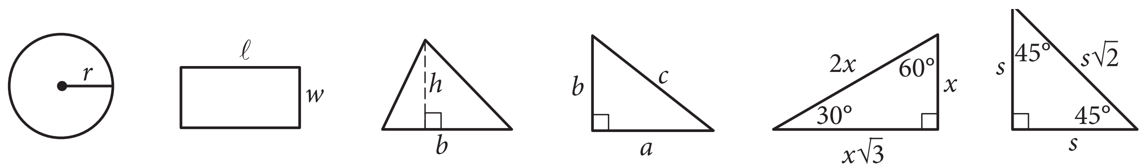
$\xrightarrow[\times 2 \text{ in } x \text{ and } y \text{ directions}]{M}$



$$A_{old} \cdot \underset{\uparrow}{4} = A_{new}$$

$\det M = 4$





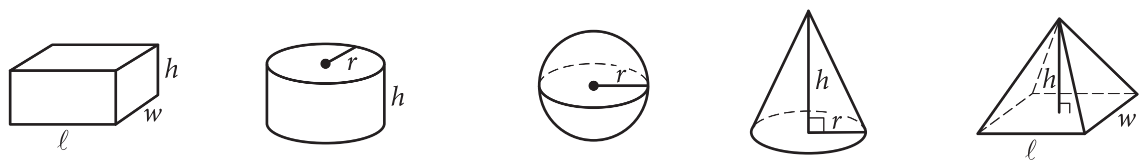
$A = \pi r^2$
$C = 2\pi r$

$A = \ell w$

$A = \frac{1}{2}bh$

$c^2 = a^2 + b^2$

Special Right Triangles



$V = \ell wh$

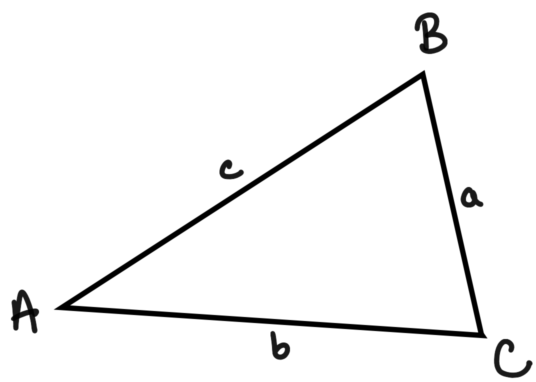
$V = \pi r^2 h$

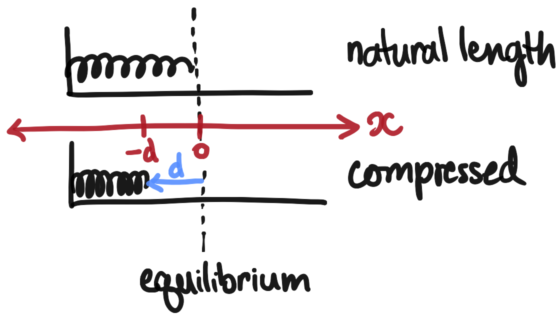
$V = \frac{4}{3}\pi r^3$

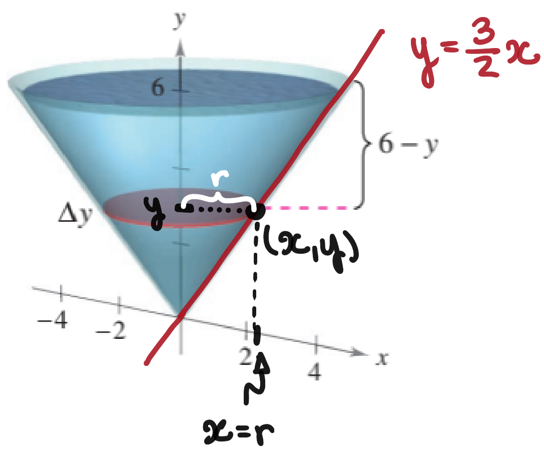
$V = \frac{1}{3}\pi r^2 h$

$V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360.
The number of radians of arc in a circle is 2π.
The sum of the measures in degrees of the angles of a triangle is 180.




natural length
x
-d
0
d
compressed
equilibrium


y
6
y = 3/2 x
6 − y
r
Δy
y
(x,y)
−4
−2
2
4
x
x=r